Detecting Writing Style Changes in Charles Dickens's Novels

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Overview

In this paper, we examine the evolution of the writing style of the author Charles Dickens. We have defined three periods of writing, with each stage containing several novels. We use supervised learning algorithms to ascertain if it is possible, through stylometric-based representation, to identify the specific stage of writing for a given text.

The features under examination include word frequency, lemmatization, part-of-speech usage, sentence length, sentiment analysis, verb tenses, named entities, punctuation marks and topics, as well as combinations thereof.

Our research question is: which features of writing most effectively differentiate between the three periods of Charles Dickens's novels?

To label the texts with their respective time periods, we initially conducted a chronological ranking of the novels based on their publication dates. Subsequently, we defined three writing stages - initial, middle, and final.

A zip file is attached comprising the following:

- Code for the project:
  - The 'main.py' file contains code for visualizations.
  - The 'data_parsing.py' file includes code for parsing the data.
  - The 'classifiers.py' file contains code to run the models.
  - The 'deviding_to_chaps.py' file contains code for dividing the books into chapters.
- Three HTML files providing descriptions of each period's topic modeling using PyLDAvis.
- An 'Input_for_AI' folder containing AI input data used for the models.
- A folder containing Charles Dickens' books.
- A folder with the books divided into chapters.

We employed the following tools for our project:

- Python for coding
- Spacy for NLP analysis and parsing data from chapters
- en_core_web_lg, the most accurate package in Spacy
- Scikit-learn for classification tasks
- Gensim for topic modeling
- PyLDAvis for visualizing topic modeling results
- NLTK for sentiment analysis
- Matplotlib for data visualization
- NumPy for numerical computations
- Pandas for data manipulation

About Charles Dickens

Charles John Huffam Dickens was an English novelist and social critic who created some of the world's best-known fictional characters. His novels and short stories are widely read today.
He was born on 7 February 1812, in Portsmouth, the second of eight children.

In 1822, the Dickens family moved to Camden Town, a poor neighborhood in London. By then, the family's financial situation had grown dire, as Charles' father had a dangerous habit of living beyond the family's means. Eventually, his father was sent to prison for debt in 1824, when Charles was just 12 years old. Dickens left school to work in a boot-blacking factory to help his family financially. These events shaped the sympathetic portrayal of orphaned and marginalized characters in Dickens's works like "Oliver Twist" and "David Copperfield". After three years he returned to school, before he began his literary career as a journalist.

In 1830, Dickens met his first love, Maria Beadnell, thought to have been the model for the character Dora in David Copperfield. Maria's parents disapproved of the courtship and ended the relationship by sending her to school in Paris.

Dickens's literary success began with the 1836 serial publication of The Pickwick Paper. Within a few years, Dickens had become an international literary celebrity, famous for his humor, satire and keen observation of character and society. His novels, most of them published in monthly or weekly installments. The installment format allowed Dickens to evaluate his audience's reaction, and he often modified his plot and character development based on such feedback.

On 2 April 1836, after a one-year engagement, and between episodes two and three of The Pickwick Papers, Dickens married Catherine Thomson Hogarth (1815–1879). They were married in London. The first of their ten children, Charles, was born in January 1837.Dickens's younger brother Frederick and Catherine's 17-year-old sister Mary Hogarth moved in with them.He became very attached to Mary, and she died in his arms after a brief illness in 1837.
Dickens idealized Mary, the character he fashioned after her, Rose Maylie, in the Oliver Twist novel, he found he could not now kill, as he had planned.

On 9 June 1865, while returning from Paris with Ellen Ternan, Dickens was involved in the Staplehurst rail crash in Kent. The train's first seven carriages plunged off a cast iron bridge that was under repair and ten passengers were killed. Dickens wasn't hurt, but he wrote less novels in the following years. He died at his house five years after the rail crash, and was buried in the Poet's Corner at Westminster Abbey.

Litrechure List

Period 1	**Period 2**	**Period 3**
The Pickwick Papers(1837) Nicholas Nickleby (1839) Oliver Twist (1839) The Old Curiosity Shop(1841) Barnaby Rudge(1841) A Christmas Carol(1843)	The Chimes(1844) Martin Chuzzlewit(1844) The Cricket on the Hearth(1845) The Battle of Life(1846) The Haunted Man(1848) Dombey and Son(1848) David Copperfield(1850) Bleak House(1853)	Hard Times(1854) Little Dorrit(1857) A Tale of Two Cities(1859) Great Expectations(1861) Our Mutual Friend (1865) The Mystery of Edwin Drood(1870)

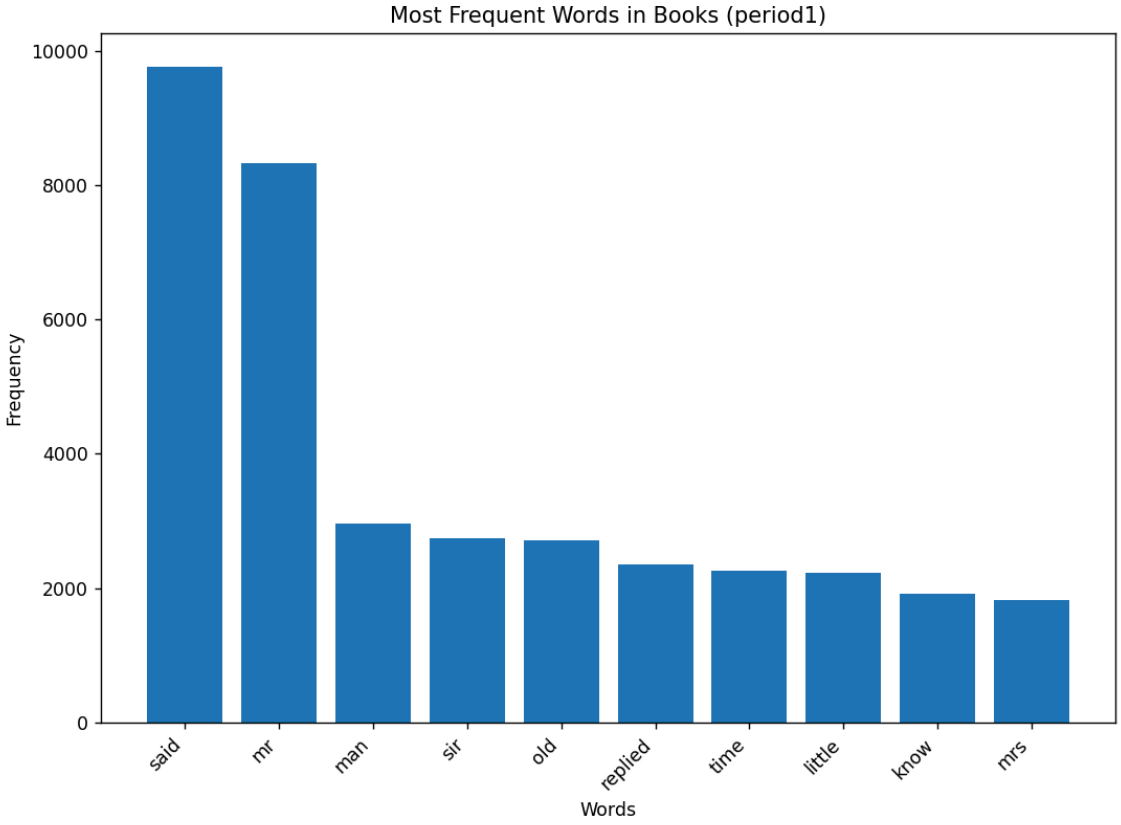
Stylometry

Stylometry is a branch of computational linguistics that focuses on quantifying and analyzing writing styles within textual data. It employs statistical and machine learning techniques to identify distinctive patterns, features, and characteristics in written texts, enabling researchers to attribute authorship, detect authorial changes, and explore writing style evolution over time. Stylometric analysis often considers factors such as vocabulary usage, sentence structure, punctuation preferences, and syntactic elements to create a unique "fingerprint" for each author or literary work. This interdisciplinary approach has wide-ranging applications, including authorship attribution in forensic linguistics, understanding historical and literary texts, and uncovering insights into the creative processes of writers and poets.

The following pages contain statistical visualizations of the three periods, including frequency of words, sentence length, frequency of lemmas, and parts of speech frequency.

To view the visualizations of topic modeling for each period please open the HTML files named "lda_visualizationperiod1", "lda_visualizationperiod2", and "lda_visualizationperiod3" from the zipped folder in your browser. These files are interactive and respond to mouse movements.

Period 1


Most Frequent Words in Books (period1)
Frequency
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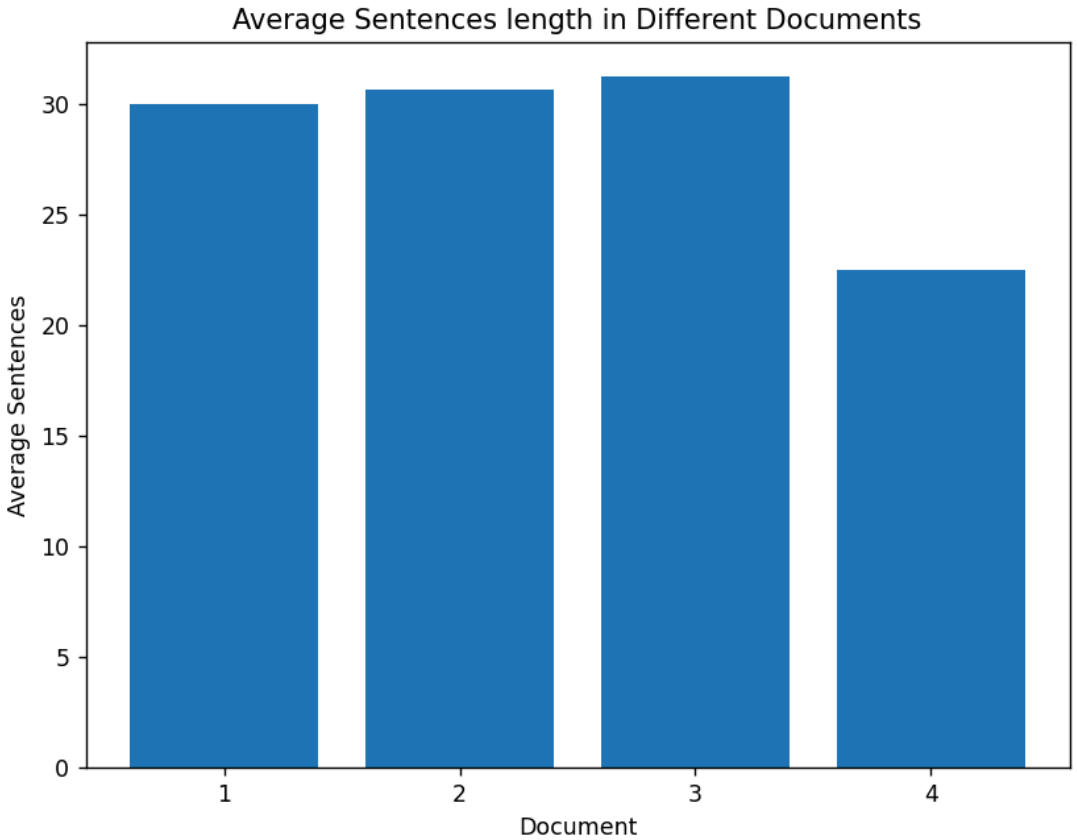
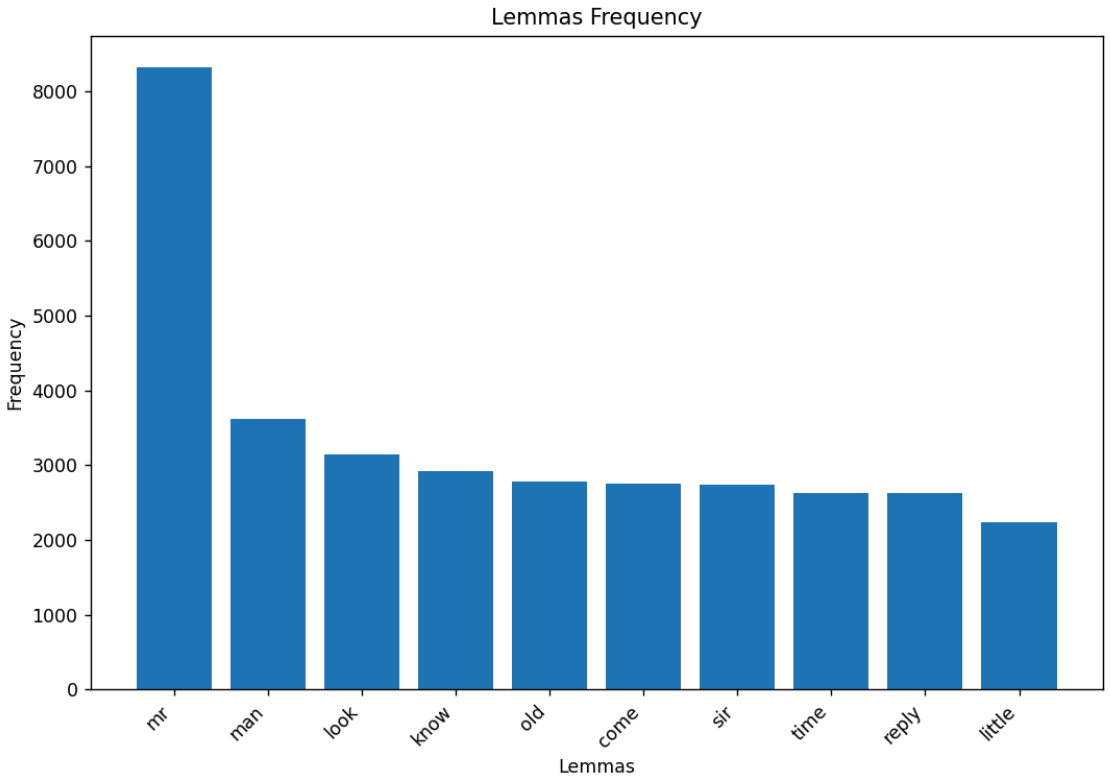
8000
6000
4000
2000
0
said
mr
man
sir
old
replied
time
little
know
mrs
Words


Average Sentences length in Different Documents
Average Sentences
30
25
20
15
10
5
0
1
2
3
4
Document


Lemmas Frequency
Frequency
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7000
6000
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3000
2000
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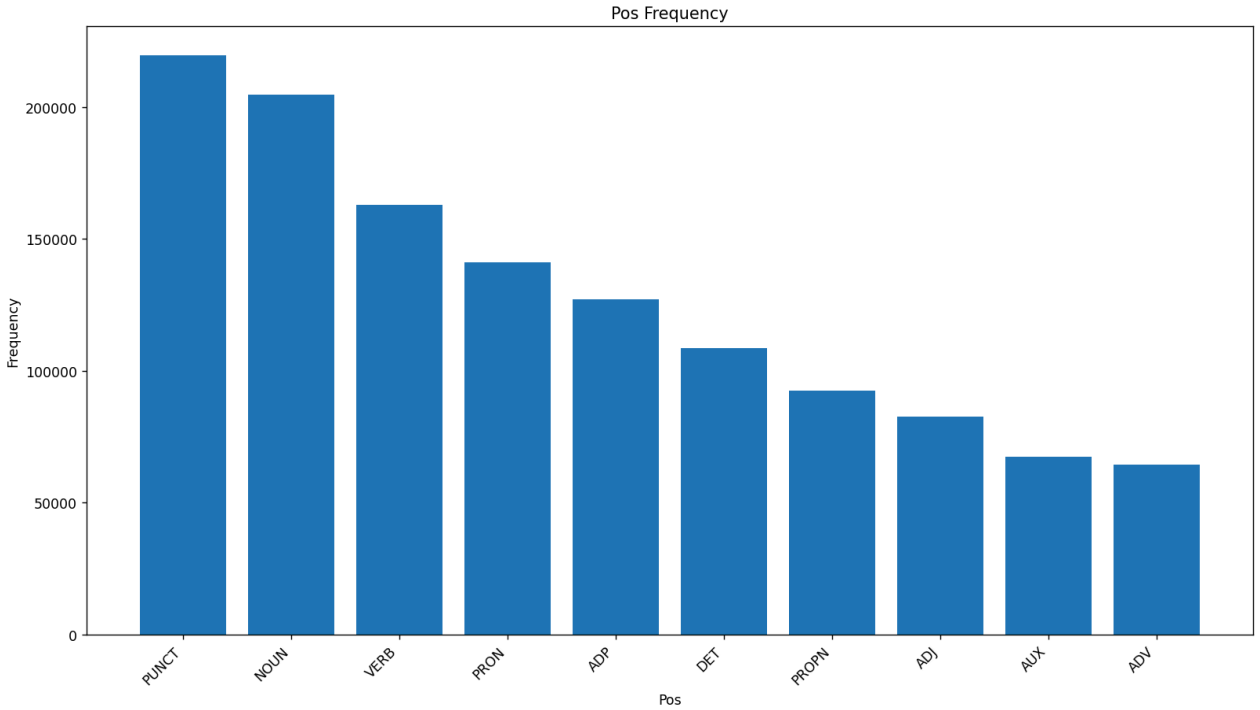
mr
man
look
know
old
come
sir
time
reply
little
Lemmas


Pos Frequency
Frequency
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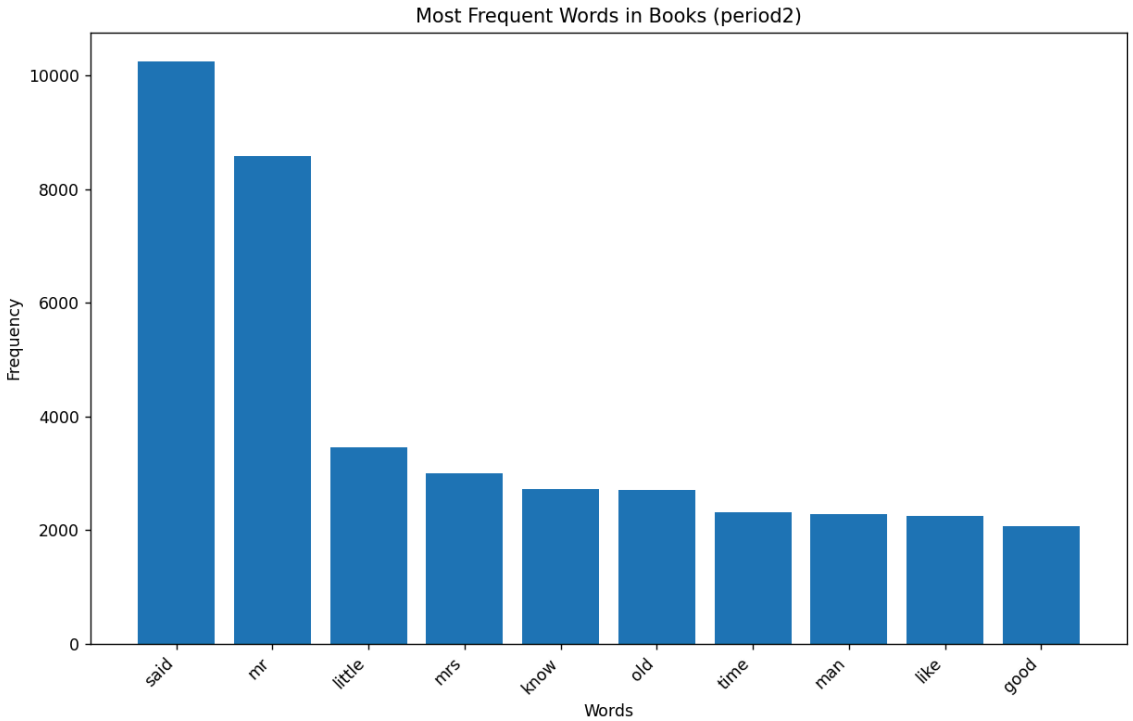
AUX
ADV
Pos

Period 2


Most Frequent Words in Books (period2)
Frequency
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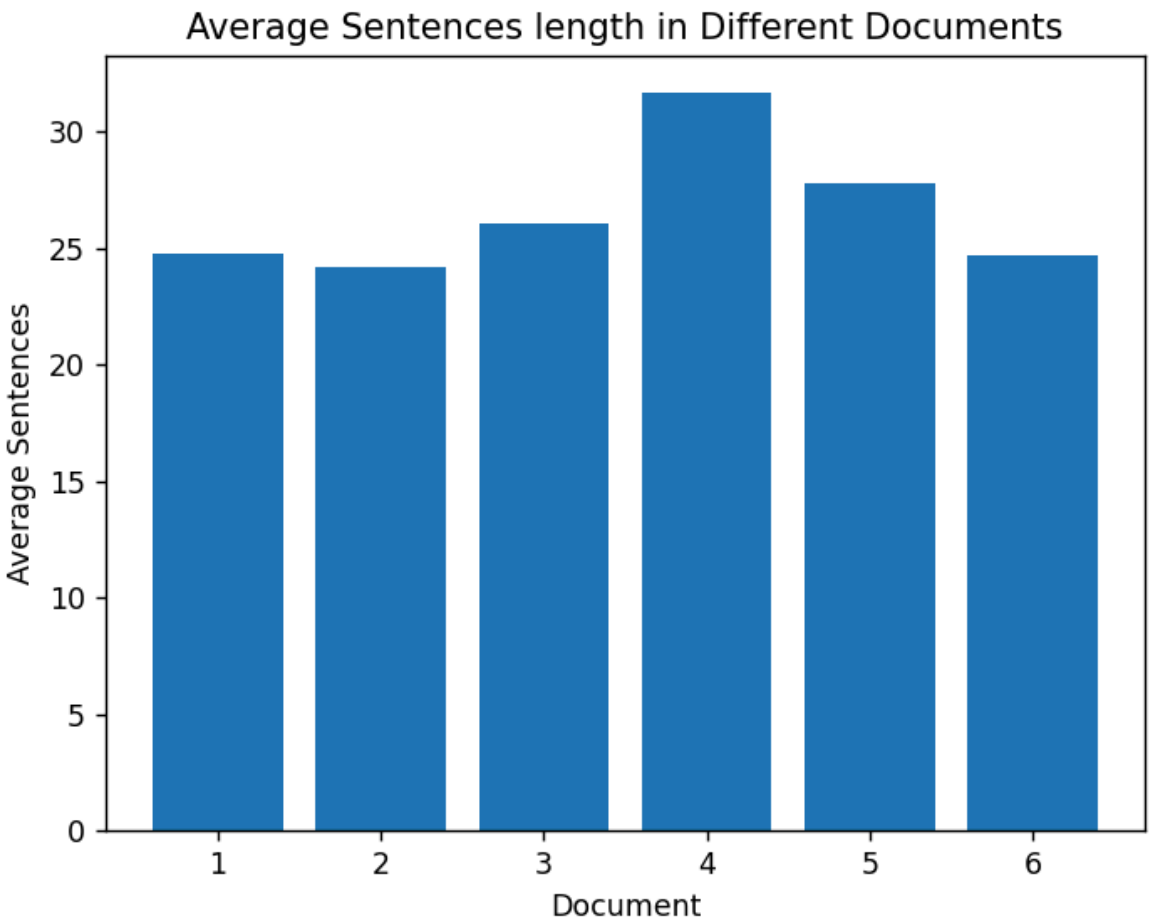
8000
6000
4000
2000
0
said
mr
little
mrs
know
old
time
man
like
good
Words


Average Sentences length in Different Documents
Average Sentences
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25
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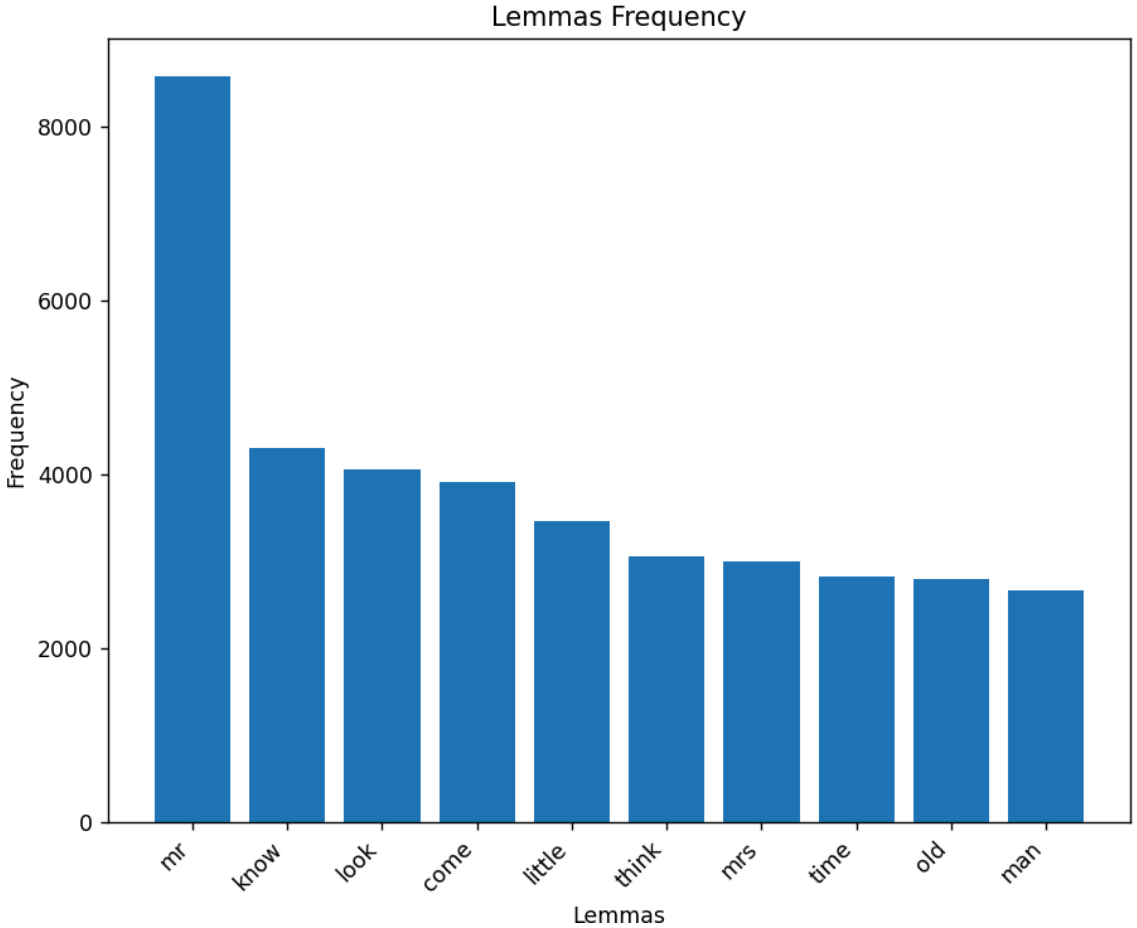
Document


Lemmas Frequency
Frequency
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6000
4000
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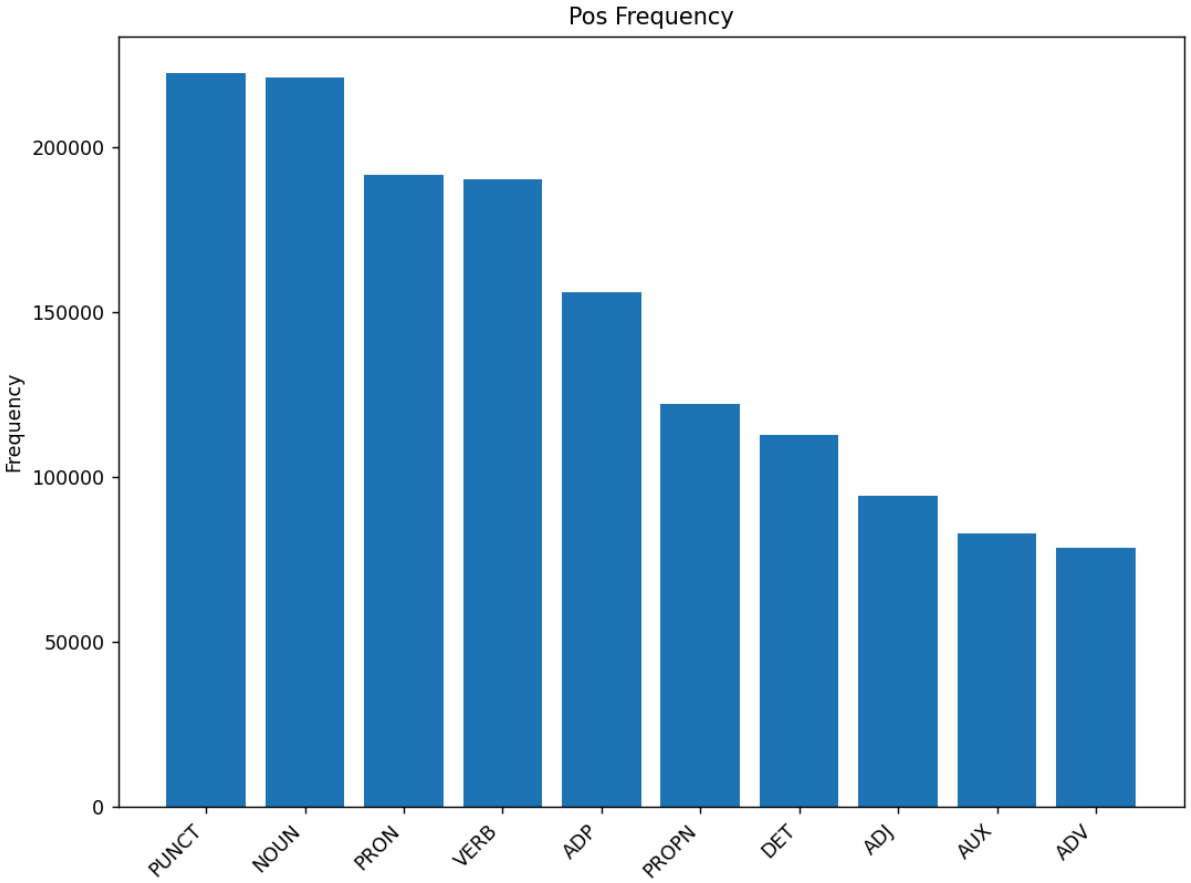
mr
know
look
come
little
think
mrs
time
old
man
Lemmas


Pos Frequency
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PUNCT
NOUN
PRON
VERB
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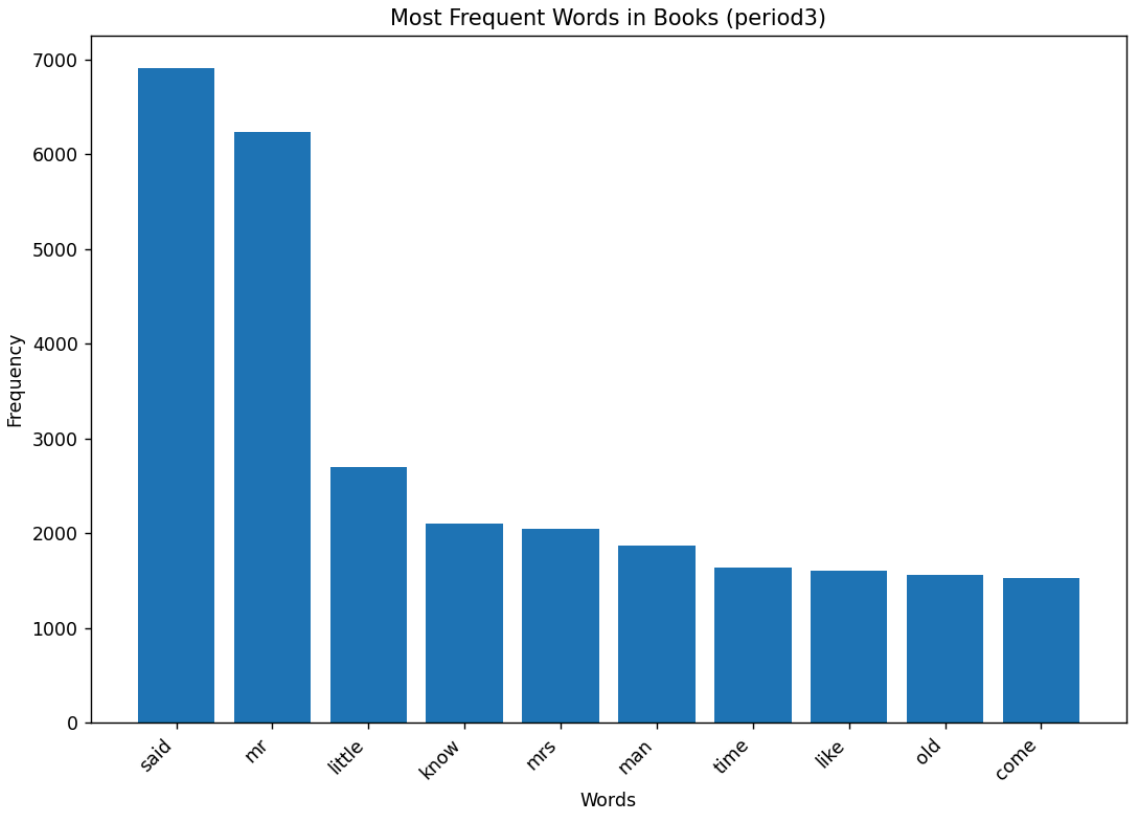
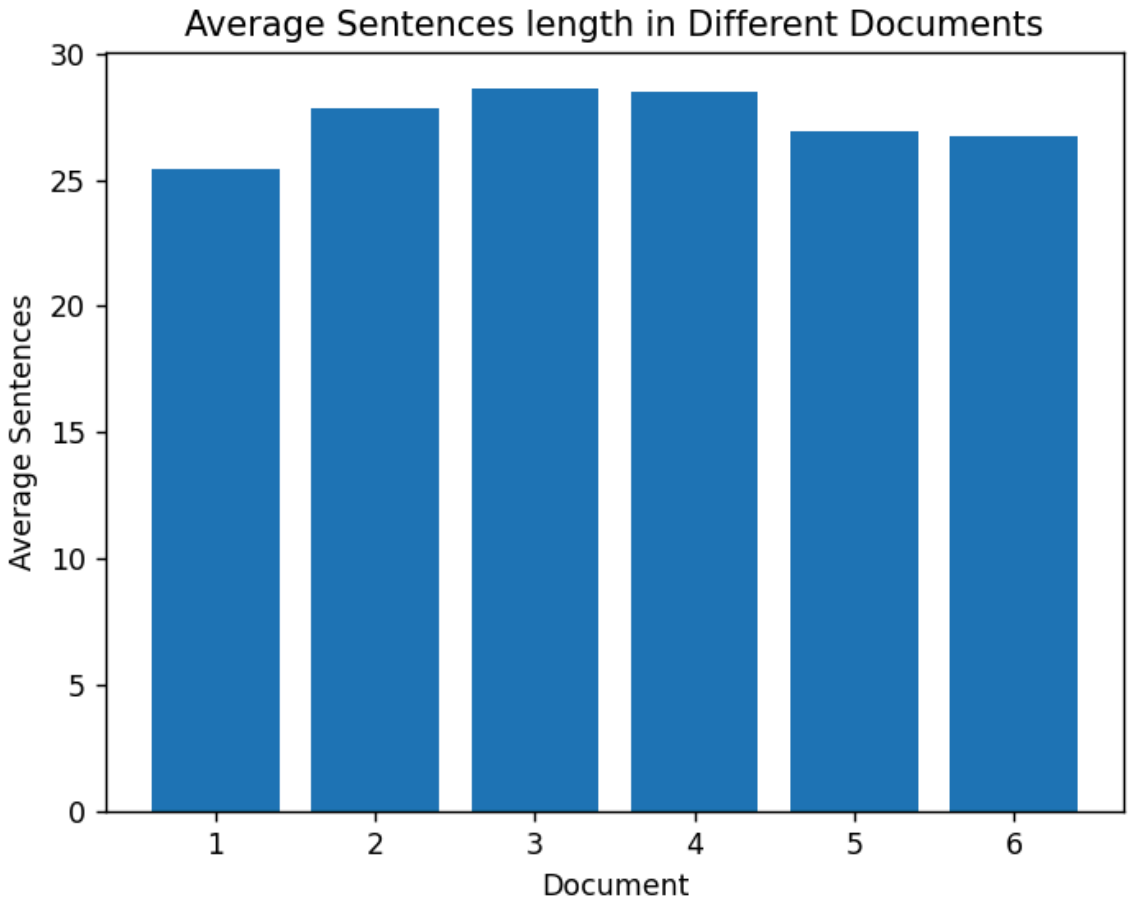
PROPN
DET
ADJ
AUX
ADV

Period 3


Most Frequent Words in Books (period3)
Frequency
7000
6000
5000
4000
3000
2000
1000
0
said
mr
little
know
mrs
man
time
like
old
come
Words


Average Sentences length in Different Documents
Average Sentences
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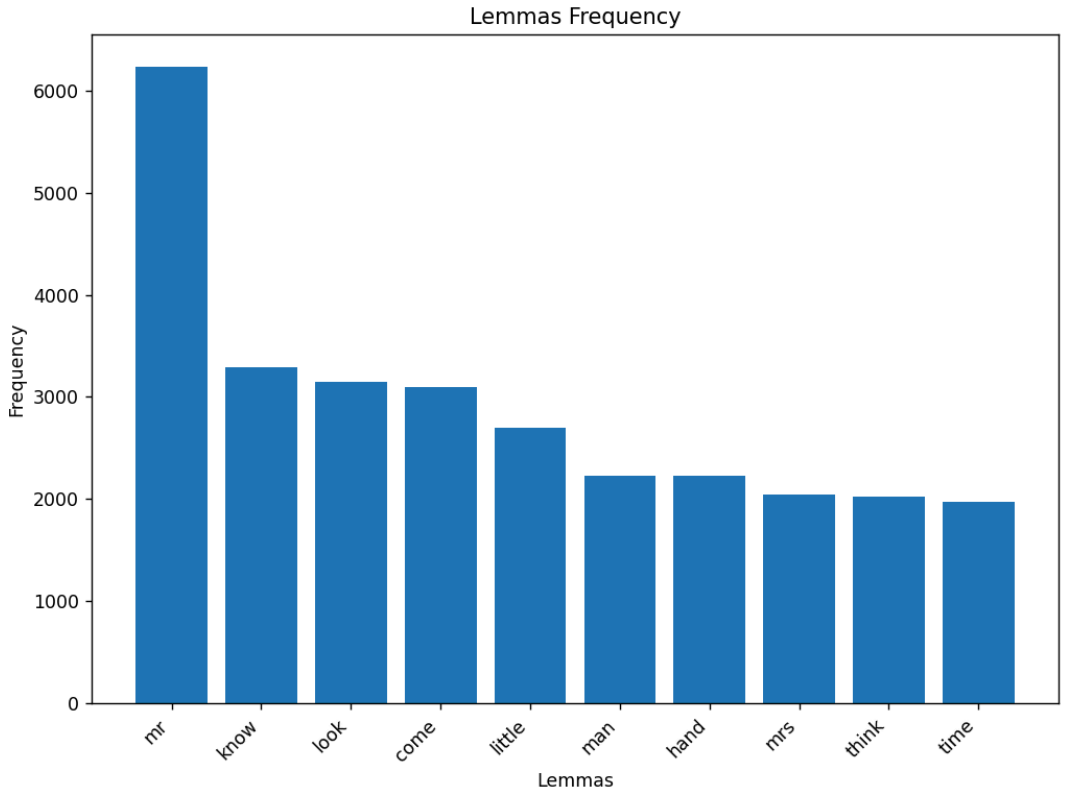
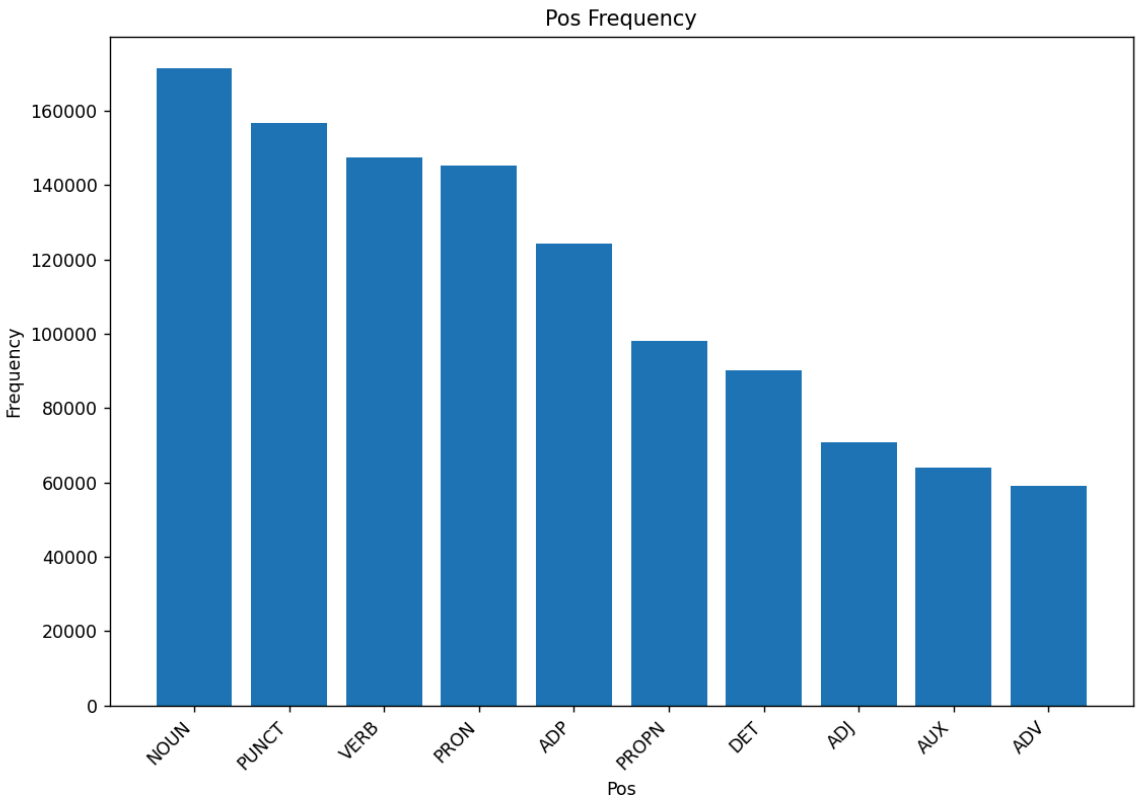
Document


Lemmas Frequency
Frequency
0
1000
2000
3000
4000
5000
6000
mr
know
look
come
little
man
hand
mrs
think
time
Lemmas


Pos Frequency
Frequency
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20000
40000
60000
80000
100000
120000
140000
160000
NOUN
PUNCT
VERB
PRON
ADP
PROPN
DET
ADJ
AUX
ADV
Pos

Visualizations Analysis

In the above presented visualizations, it is evident that certain words are consistently prominent across all periods analyzed. Specifically, the terms "said" and "mr" stand out as the most frequently utilized words. However, an intriguing observation emerges when examining the prevalence of the word "little" in period 1, indicating its unique significance within this subset of texts.

Moreover, an examination of the part-of-speech (POS) distribution reveals noteworthy trends. Both periods 1 and 2 exhibit a preponderance of punctuation marks, while period 3 displays a distinct emphasis on nouns. Conversely, adverbs appear to be the least utilized POS category across all periods studied.

Regarding sentence structure, the average sentence length is approximately 30 words. However, a notable deviation is observed in period 2, where the average sentence length tends to hover closer to 25 words.

Conclusions Based on the findings presented:

Consistent Word Usage: The frequent appearance of certain words such as "said" and "mr" across all periods suggests their integral role in the author's narrative style and dialogue construction.

Unique Word Patterns: The prevalence of the word "little" in period 1, distinct from other periods, may indicate thematic or stylistic nuances specific to the literary works within this timeframe.

Furthermore, it's noteworthy that the term "Mrs." appears most frequently during period 2, which encompasses the story "David Copperfield." In this celebrated novel, Dickens draws inspiration from his own romantic history, as he models the character of Dora after his first love, Maria Beadnel.

Part-of-Speech Patterns: The dominance of punctuation in periods 1 and 2, followed by a shift towards noun usage in period 3, highlights potential shifts in narrative focus or linguistic expression over time.

Sentence Structure: The observed differences in average sentence length between periods, particularly the shorter average length in period 2, may reflect evolving narrative techniques or stylistic preferences adopted by the author.

Exploring Research Question- Classification of Periods by Features

To evaluate our research question, we employed classifiers available in the Scikit-Learn library within Python. The selected features for analysis included:

1. The proportion of parts of speech (utilizing the SpaCy library), such as verbs, adjectives, adverbs, interjections, punctuation, and nouns.
2. The percentage of frequent words.
3.The percentage of frequent lemmas.
4. Sentiment analysis scores (leveraging the NLTK sentiment analysis tool) for negative, positive, and neutral sentiments.
5. Percentage distribution of verb tenses.
6. Percentage of named entities.
7. Analysis of punctuation marks, specifically question marks and exclamation marks, to provide a more nuanced understanding of punctuation usage.

We conducted the classification analysis at the chapter level. This involved dividing each of Charles Dickens' books into chapters and then extracting features from each chapter for further analysis. We undertook this approach to ensure a sufficient amount of input data for the models.

To determine which writing feature most effectively distinguishes between the three periods, we conducted several analyses using classifiers. First, we began by using a matrix that incorporated all seven features. Additionally, we performed separate classifier runs for each feature, systematically excluding one feature at a time from the list of seven. This approach allowed us to assess the impact of each individual feature by observing how the classifier performed when one feature was omitted from the analysis.

Our analysis involved three models: MLP classifier, decision tree, and logistic regression. By employing these diverse models, we aimed to capture different aspects of feature importance and classification accuracy.

Furthermore, we ran the classifiers on each feature in isolation to evaluate their individual effectiveness. This step is aimed to confirm our initial hypotheses regarding which feature might have the most significant impact on distinguishing between the periods. We expected that a feature identified as highly impactful in the initial analysis would also demonstrate the highest classification accuracy when evaluated independently compared to other features.

The running classification results (accuracy scores):

<u>MLP:</u>

all features: 0.6919642857142857
all features excluding punctuation marks: 0.6294642857142857
all features excluding frequent words: 0.6428571428571429
all features excluding sentiments: 0.6428571428571429
all features excluding verb tenses: 0.7366071428571429
all features excluding named entities: 0.71875
all features excluding pos: 0.6026785714285714
all features excluding frequent lemmas: 0.71875
pos: 0.5625
frequent words: 0.42857142857142855
frequent lemmas: 0.36607142857142855

sentiments: 0.4732142857142857
verb tenses: 0.3705357142857143
named entities: 0.3705357142857143
punctuation marks: 0.4642857142857143

Decision Tree:

all features: 0.6026785714285714
all features excluding punctuation marks: 0.6205357142857143
all features excluding frequent words: 0.59375
all features excluding sentiments: 0.5848214285714286
all features excluding verb tenses: 0.5625
all features excluding named entities: 0.5625
all features excluding pos: 0.5625
all features excluding frequent lemmas: 0.6428571428571429
pos: 0.5
frequent words: 0.46875
frequent lemmas: 0.35267857142857145
sentiments: 0.3794642857142857
verb tenses: 0.34375
named entities: 0.29017857142857145
punctuation marks: 0.41517857142857145

Logistic Regression:

all features: 0.7544642857142857
all features excluding punctuation marks: 0.6919642857142857
all features excluding frequent words: 0.7321428571428571
all features excluding sentiments: 0.7053571428571429
all features excluding verb tenses: 0.75
all features excluding named entities: 0.7544642857142857
all features excluding pos: 0.6696428571428571
all features excluding frequent lemmas: 0.78125
pos: 0.6205357142857143
frequent words: 0.42857142857142855
frequent lemmas: 0.39285714285714285
sentiments: 0.4732142857142857
verb tenses: 0.4017857142857143
named entities: 0.3705357142857143
punctuation marks: 0.5580357142857143

Analysis of the results

In the MLP model, the most impactful feature was the percentage of parts of speech (POS), excluding it results in a decrease of 9% in accuracy. When evaluated alone, POS had the highest score of 56%. The second most influential feature was punctuation marks, where excluding it led to a decrease in accuracy of 7%, and as a standalone feature, it achieved a relatively high score of 46%.

On the other hand, excluding verb tenses percentage resulted in a 4% increase in accuracy, while excluding named entities and frequent lemmas increased accuracy by 2% each. This indicates that these latter features may introduce noise or confusion to the model's predictions.

In the decision tree model, the most impactful feature was the percentage of parts of speech (POS). Excluding POS led to a decrease in accuracy of 4%, which was also observed when excluding verb tenses percentage or named entities. As an individual feature, POS achieved the highest score of 50%. However, verb tenses and named entities did not perform as well as standalone features.

Additionally, excluding frequent lemmas percentage resulted in an increase of 4% in accuracy, while excluding punctuation marks increased accuracy by 2%. This suggests that these particular features may introduce noise or disrupt the model's predictions.

In logistic regression, the most impactful feature was the percentage of parts of speech (POS). Excluding POS led to a decrease in accuracy of 9%, and as an individual feature, POS achieved the highest score of 62%. The second most effective feature was punctuation marks, with a decrease in accuracy of 6% when excluded, and as a standalone feature, it achieved a relatively high score of 55%.

Furthermore, excluding frequent lemmas percentage led to an increase of 3% in accuracy. This suggests that frequent lemmas may introduce noise or disrupt the model's predictions.

Upon analyzing the results, we conducted another round of model runs excluding the noisy features identified earlier.

The updated results:

MLP: 0.7366071428571429
Decision Tree: 0.6026785714285714
Logistic Regression: 0.78125

We observed improvements in accuracy for both MLP and logistic regression models after removing the noisy features. However, the decision tree model produced the same result, indicating that the removal of noisy features did not lead to an improvement in accuracy for this particular model.

The results reveal notable distinctions in model performance, with logistic regression emerging as the most accurate, followed by MLP and then the decision tree model, which exhibited the lowest accuracy. This disparity underscores the trade-off between model complexity and performance, where logistic regression's simplicity contributed to its superior accuracy. Moreover, the importance of features varied across models, with logistic regression and MLP highlighting the significance of features such as POS percentage and punctuation marks. The relatively stable performance of the decision tree model, even after excluding noisy features, suggests potential limitations in capturing nuanced patterns in the data. Logistic regression's strong performance not only indicates its suitability for this specific writing style analysis task but also emphasizes its interpretability, which facilitates understanding the factors influencing the model's predictions.

Conclusions on the Research Question

After analyzing the results, it becomes evident across all models that the most impactful feature is the POS (Parts of Speech) percentage. This feature captures Charles Dickens' writing style evolution across the three periods, highlighting the relative usage of verbs, adjectives, adverbs, interjections, punctuation, and nouns.

In both MLP and logistic regression models, the usage of question marks and exclamation marks also emerged as effective features, shedding light on Charles Dickens' writing style variations over time.

Additionally, it is notable that frequent lemmas consistently disrupted the model's performance in all models, suggesting that they do not accurately characterize the changes in his writing style.

Dickens' writing style is intricately connected to his background as a journalist and astute observer of society. This influence is notably reflected in his adept use of different parts of speech, a feature highlighted as the most impactful in the classifiers' results. Dickens' skillful manipulation of verbs, adjectives, adverbs, and other linguistic components allows him to craft vivid descriptions, evoke emotions, and develop multifaceted characters within his narratives. Moreover, his strategic use of punctuation marks, particularly question marks and exclamation marks, further accentuates his narrative style and penchant for dramatic storytelling. The classifiers' recognition of the effectiveness of these punctuation marks aligns with Dickens' deliberate employment of rhetorical questions, exclamatory expressions, and dialogue to engage readers and heighten the dramatic tension in his works.

In conclusion, the analysis underscores the importance of the linguistic features of POS percentages and punctuation usage in capturing the nuances of Charles Dickens' writing style evolution. The consistent disruption caused by frequent lemmas highlights the need for careful feature selection in NLP tasks. These findings not only contribute to understanding Dickens's literary journey but also emphasize the significance of computational techniques in unraveling intricate patterns within textual data. Additionally, it's worth noting that logistic regression achieved the highest accuracy score of 78% in the analysis, further showcasing the effectiveness of certain models in handling textual data.

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